Rohan Reddy Thumma

US Citizen • Florham Park, NJ • (914) 525-9546
LinkedIn • rrthumma29@yahoo.com • GitHub

Education

Rensselaer Polytechnic Institute (RPI), *Troy, NY* *August 2021 – May 2026*
BS in Computational Biology | Lally School of Management MBA, AI and Advanced Computing — *GPA: 3.7*

Skills

Programming: Python (Pandas, NumPy, Scikit-learn), JavaScript, R, SQL/T-SQL
Frameworks & APIs: FastAPI, React/Vite, RESTful services, Node.js (exosure), Flask/Django (exposure)
Databases: PostgreSQL, MySQL, Azure SQL, MongoDB, BigQuery, SQL Server (SSMS)
Cloud & DevOps: AWS (EC2, S3, Lambda, IAM, VPC), Azure, GCP, Git, GitHub, Linux/Ubuntu, SSH
Dev Tools: VS Code, pgAdmin, Power BI, Tableau, RStudio, PuTTY, Git

Professional Experience

Rensselaer Polytechnic Institute, *Troy, NY* *September 2025 – Present*
Master's Course Assistant – Data Science / Xinformatics

- Supported instruction on data architecture, metadata standards (METS, ESML, netCDF), and reproducible scientific workflows, focusing on lifecycle-aware data management and scalable research data pipelines.
- Evaluated Python assignments implementing ML pipelines, ETL workflows, metadata encoding, and provenance tracking, reinforcing best practices in data engineering, pipeline design, and reproducible computing systems.

Alithya, *Indianapolis, IN (Remote)* *October 2024 – August 2025*
System Analyst

- Engineered cloud-native ETL pipelines and queue orchestration systems using Power Automate and Azure SQL, leveraging Microsoft Copilot for prompt-driven workflow generation and validation —reducing processing errors by 20%
- Architected scalable analytics infrastructure and Power BI reporting layers to surface claim-error patterns across billing workflows, utilizing Copilot-assisted DAX query generation— reducing manual reconciliation time by 40%.

Motivity Labs/Walmart, *Irving, TX* *May 2024 – August 2024*
Data Analytics Intern

- Engineered secure cloud ingestion pipelines integrating enterprise server infrastructure with Google BigQuery, processing 100GB+ daily call-center datasets to support distributed analytics and scalable operational reporting.
- Architected Tableau analytics infrastructure and optimized warehouse data models with automated refresh pipelines and query performance tuning — reducing manual reporting workloads by 50% and enabling real-time KPI visibility.

Projects

NutriTrack – Full-Stack Social Nutrition & Gamification App, *Troy, NY* *March 2026 – Present*

- Architected a 13-table relational schema and production-grade REST API using FastAPI and SQLAlchemy 2.0 — implementing JWT authentication, Pydantic validation, and 51 endpoints across 7 modular routers.
- Engineered background task processing for a gamification engine covering XP awards, streak tracking, level progression, and 17 achievements — with an idempotent XP ledger and 92 USDA-sourced seeded foods.
- Built end-to-end via agentic AI-assisted development (Claude Code in VS Code) — delivering auth, food logging, gamification, and a social feed in a Dockerized app verified by a 30-assertion smoke test.

Lally Master's Scholars Research Program— Fiduciary AI Trust Platform, *Troy, NY* *January 2026 – Present*

- Engineered a 3-condition between-subjects experiment comparing human vs. AI advisor trust — built full stimulus suite (branded website, Synthesia avatar videos, Qualtrics behavioral instrumentation) using agentic AI development.
- Piloting a perception manipulation-check before deploying to 200+ students, with a follow-on study targeting wealth management professionals; owning full lifecycle from experimental design through technical delivery.

Sub-Saharan Cholera Analysis: GDP & Water Sanitation, *Troy, NY* *September 2023 – March 2024*

- Wrangled 15+ years of multi-format public health data (netCDF precipitation, World Bank GDP, shapefile geospatial) using Python (pandas, scipy), resolving format incompatibilities via custom HDF5-to-CSV conversion scripts.
- Applied scikit-learn K-Means and Mean Shift clustering alongside Spearman regression (~0.70) to model cholera outbreak drivers; visualized findings geospatially with geopandas / matplotlib
- Co-authored and presented an ACM ICMHI publication in Yokohama, Japan — DOI: 10.1145/3673971.3673996

Certifications & Extracurriculars

Preparing: AWS Certified Cloud Practitioner | AWS Certified Machine Learning

Pi Delta Psi Cultural Fraternity | RPI Volleyball | Society of Asian Scientists and Engineers | RPI Blockchain club